Music Store Analysis

Practice Questions & Solutions

Q1. Who is the senior most employee based on job title?

Solution:

```
SELECT
    employee_id,
    CONCAT_WS(' ',first_name,last_name),
    title,
    levels,
    phone,
    email
FROM employee
WHERE levels = (SELECT MAX(levels) FROM employee);
```

Q2. Which countries have the most Invoices?

Solution:

```
SELECT
    billing_country,
    COUNT(invoice_id) AS inv
FROM invoice
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1;
```

Q3. What are the top 3 values of total invoice?

Solution:

```
SELECT
    i.customer_id,
    CONCAT_WS(' ',c.first_name,c.last_name) AS customer_name,
    SUM(i.total) AS total
FROM invoice i JOIN customer c ON i.customer_id = c.customer_id
GROUP BY 1,2
ORDER BY 3 DESC
```

```sql
LIMIT 3;
```

Q4. Which city has the best customers? We would like to throw a promotional Music Festival in the city we made the most money.

Write a query that returns one city that has the highest sum of invoice totals. Return the city with the highest sum of invoice totals along with the total amount.

Solution:

```sql
SELECT
    billing_city,
    SUM(total) AS total
FROM invoice
GROUP BY 1
ORDER BY 2 DESC
LIMIT 1;
```

Q5. Who is the best customer? The customer who has spent the most money will be declared the best customer.

Write a query that returns the person who has spent the most money.

Solution:

```sql
SELECT
    i.customer_id,
    CONCAT_WS(' ',c.first_name,c.last_name) AS customer_name,
    SUM(i.total) AS total
FROM invoice i JOIN customer c ON i.customer_id = c.customer_id
GROUP BY 1,2
ORDER BY 3 DESC
LIMIT 1;
```

Q6. Write query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A

Solution:

```sql
SELECT DISTINCT
    CONCAT_WS(' ',c.first_name,c.last_name) AS customer_name,
    c.phone,
    c.email,
    g.name AS genre
FROM customer c JOIN invoice i      ON i.customer_id = c.customer_id
```

```sql
                JOIN invoice_line il ON i.invoice_id  = il.invoice_id
                JOIN track t         ON t.track_id    = il.track_id
                JOIN genre g         ON g.genre_id    = t.genre_id
WHERE g.name IN ('Rock')
ORDER BY 1 ASC;
```

Q7. Lets invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands

Solution:

```sql
SELECT
    a.artist_id,
    a.name AS artist_name,
    COUNT(t.track_id) AS total_track_count
FROM artist a JOIN album alb ON a.artist_id  = alb.artist_id
              JOIN track t   ON alb.album_id = t.album_id
              JOIN genre g   ON g.genre_id   = t.genre_id
WHERE g.name IN ('Rock')
GROUP BY 1,2
ORDER BY 3 DESC
LIMIT 10;
```

Q8. Return all track names that have a song length longer than the average song length, ordered by longest first. Return the Name and Milliseconds for each track. Order by the song length with the longest songs listed first.

Solution:

```sql
SELECT
    name,
    milliseconds
FROM track
WHERE milliseconds > (SELECT AVG(milliseconds) FROM track)
ORDER BY 2 DESC;
```

Q9. Find how much amount each customer spent on artists. Return customer name, artist name and total spent.

Solution:

```sql
SELECT
    CONCAT_WS(' ',c.first_name,c.last_name) AS customer_name,
```

```
    a.name AS artist_name,
    SUM(il.unit_price * il.quantity) AS total_spend
FROM customer c JOIN invoice i       ON i.customer_id = c.customer_id
                JOIN invoice_line il ON i.invoice_id  = il.invoice_id
                JOIN track t         ON t.track_id    = il.track_id
                JOIN album alb       ON alb.album_id  = t.album_id
                JOIN artist a        ON a.artist_id   = alb.artist_id
GROUP BY 1,2
ORDER BY 1,3 DESC;
```

Q10. We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared return all Genres.

Solution:

```
WITH top_genre AS
(
SELECT
    i.billing_country AS country,
    g.name AS genre_name,
    COUNT(g.genre_id) AS no_of_purchases
FROM invoice i  JOIN invoice_line il ON i.invoice_id  = il.invoice_id
                JOIN track t         ON t.track_id    = il.track_id
                JOIN genre g         ON g.genre_id    = t.genre_id
GROUP BY 1,2
),
ranks AS
(
SELECT
    *,
    DENSE_RANK() OVER(PARTITION BY country ORDER BY no_of_purchases
DESC) AS rno
FROM top_genre
)
SELECT
    *
```

```sql
FROM ranks
WHERE rno IN (1);
```

Q11. Determine the customer that has spent the most on music for each country. For countries where the top amount is shared, return all customers.

Solution:

```sql
WITH top_customer AS
(
SELECT
    c.country,
    CONCAT_WS(' ',c.first_name,c.last_name) AS customer_name,
    SUM(i.total) AS total_spent
FROM customer c JOIN invoice i ON i.customer_id = c.customer_id
GROUP BY 1,2
),
ranks AS
(
SELECT
    *,
    DENSE_RANK() OVER(PARTITION BY country ORDER BY total_spent DESC)
AS rno
FROM top_customer
)
SELECT
    *
FROM ranks
WHERE rno = 1;
```